CPSC-406 Report

Angelina House
Chapman University

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Abstract

Contents

1 Introduction **1**

2 Week by Week **1**
2.1 Week 1 1
2.2 Week 2 2
2.3 Week 3 2

3 Synthesis **3**

4 Evidence of Participation **3**

5 Conclusion **3**

1 Introduction

2 Week by Week

2.1 Week 1

Notes

Week One Notes.

Homework

None

Reflection

comments and questions

2.2 Week 2

Notes

This week we covered Derterministic Finite Automata (DFA) and continued to build on our definition on automata from week one.

Homework

Question 1: Determine for the following words, if they are contained in L1, L2, or L3

The equations are:

$$L_1 := \{x01y \mid x, y \in \Sigma^*\}$$
$$L_2 := \{w \mid |w| = 2^n \text{ for some } n \in \mathbb{N}\}$$
$$L_3 := \{w \mid |w|_0 = |w|_1\}$$

Using the given chart and the given equations we can come to the following conclusions.

Word	L_1 (Contains "01")	L_2 (Length 2^n)	L_3 (Equal 0s and 1s)
$w_1 = 10011$	✓	x	x
$w_2 = 100$	x	x	x
$w_3 = 10100100$	✓	✓	×
$w_4 = 1010011100$	✓	×	✓
$w_5 = 11110000$	×	✓	✓

By following the the language equations, conditions, and comparing using the word we can determine whether the word would be accepted in the equation.

Question 2: Let Σ be some fixed alphabet and $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ be a finite automaton. The DFA $\mathcal{A}$ can "process" a word by tracing through it via a sequence of transitions corresponding to the letters of the word. Consider the paths corresponding to the words:

$$w_1 = 0010, \quad w_2 = 1101, \quad \text{and} \quad w_3 = 1100.$$

For which of these words does their run end in the accepting state?

w_1 and w_2 end in the accepting state. Following the given graph and going through the word we can determine that w_1 and

Reflection

comments and questions

2.3 Week 3

Week Three notes.

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, Title, Publisher, Year.